

MILWAUKEE MITCHELL, WI, USA

WMO: 726400

Lat: 42.955N

Lon: 87.904W

Elev: 670

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	2.8	-13.6	2.7	-0.2	-8.9	3.5	3.9	26.5	25.3	24.3	26.4	11.7	290	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	89.5	74.3	86.4	72.3	83.5	70.6	76.7	86.2	74.8	83.2	73.0	80.7	13.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	127.8	82.1	71.9	120.7	80.2	70.2	114.0	78.2	40.4	86.2	38.6	83.4	37.0	80.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	21.0	19.2	DB	-8.1	94.6	6.8	3.7	-13.0	97.3	-17.0	99.4	-20.8	101.5	-25.7	104.2	(4)
			WB	-9.0	79.6	6.5	2.3	-13.7	81.3	-17.5	82.7	-21.2	84.0	-26.0	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	23.0	26.2	35.6	45.7	56.2	66.9	72.6	71.6	64.6	52.6	39.7	28.8	(6)
		DBStd	19.41	11.40	10.74	10.38	8.59	8.80	8.06	6.16	5.33	7.50	8.42	9.36	10.30	(7)
		HDD50	3241	836	668	463	184	33	0	0	2	69	327	659		(8)
		HDD65	6647	1301	1087	911	583	301	74	9	98	393	758	1123		(9)
		CDD50	2784	0	1	18	55	226	507	699	668	439	151	19	1	(10)
		CDD65	715	0	0	1	3	29	131	243	212	86	10	0	0	(11)
		CDH74	5231	0	0	6	55	272	1080	1945	1277	532	64	0	0	(12)
		CDH80	1548	0	0	0	11	63	349	645	339	131	10	0	0	(13)
		WSAvg	9.7	10.7	10.6	10.5	11.1	9.9	8.5	8.5	8.0	8.6	9.8	10.2	10.1	(14)
		PrecAvg	32.9	1.8	1.5	2.4	3.6	3.1	3.9	3.4	3.5	3.1	2.4	2.2	2.0	(15)
(16)	Precipitation	PrecMax	45.1	4.4	3.5	6.9	7.4	8.4	12.3	10.9	9.1	9.4	7.0	5.4	5.4	(16)
(17)		PrecMin	19.1	0.3	0.0	0.3	1.3	0.5	0.7	0.7	0.6	0.0	0.2	0.4	0.3	(17)
(18)		PrecStd	5.8	1.0	0.9	1.4	1.5	1.7	2.3	2.0	2.0	2.1	1.3	1.1	1.2	(18)
(19)																
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	52.2	58.0	72.9	81.2	86.2	92.7	95.1	91.3	88.5	81.1	67.7	57.9		(19)
(21)		0.4% MCWB	47.9	50.4	59.8	62.1	68.9	75.3	77.1	75.9	72.3	67.5	57.7	54.2		(20)
(22)		2% DB	45.4	49.6	63.2	72.8	81.1	88.1	90.0	87.1	83.7	74.7	61.9	50.5		(21)
(23)		2% MCWB	41.0	43.3	54.5	58.9	66.3	72.4	75.5	73.4	70.1	63.9	54.0	46.3		(22)
(24)		5% DB	40.3	43.4	56.4	66.0	76.5	84.0	86.5	83.6	80.1	70.4	57.8	45.6		(23)
(25)		5% MCWB	37.3	38.7	48.5	54.3	64.1	70.5	73.2	71.4	68.4	61.2	51.6	42.2		(24)
(26)		10% DB	37.5	39.4	50.1	60.2	71.8	80.2	83.3	80.7	76.3	65.7	53.2	41.8		(25)
(26)		10% MCWB	35.3	35.9	43.7	50.5	61.4	68.5	71.3	70.1	67.1	58.5	47.6	38.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	49.5	51.3	62.0	65.3	72.0	77.6	80.2	77.9	75.1	70.5	59.6	54.5		(27)
(28)		0.4% MCDB	51.1	55.6	71.5	77.3	82.5	88.7	90.3	87.7	84.5	78.4	64.5	57.6		(28)
(29)		2% WB	41.4	44.0	56.2	60.0	68.5	74.7	77.3	75.8	72.4	66.4	56.1	47.4		(29)
(30)		2% MCDB	44.6	48.9	62.5	71.1	77.7	85.0	87.1	83.5	80.1	71.9	60.7	49.5		(30)
(31)		5% WB	37.7	39.4	49.4	55.9	65.8	72.2	75.2	73.8	70.6	62.7	51.8	42.6		(31)
(32)		5% MCDB	40.0	43.2	55.4	65.0	74.3	81.9	83.9	80.7	77.4	68.4	56.5	45.1		(32)
(33)		10% WB	35.1	36.2	43.9	51.0	62.6	69.9	73.3	72.1	68.6	59.5	48.1	38.7		(33)
(34)		10% MCDB	37.5	39.2	49.5	58.7	70.6	77.8	81.0	78.2	74.8	65.3	52.6	41.5		(34)
(35)	Mean Daily Temperature Range	MDBR	12.5	12.7	13.9	15.4	16.7	16.5	15.4	14.1	15.4	14.7	13.1	11.7		(35)
(36)		5% DB MCDBR	15.2	17.5	22.8	27.0	23.8	20.8	18.8	17.8	19.7	21.1	19.0	15.7		(36)
(37)		5% DB MCWBR	13.2	13.2	16.1	16.4	13.8	10.9	9.2	8.7	9.7	12.9	14.1	13.1		(37)
(38)		5% WB MCDBR	14.0	16.6	21.6	25.8	22.4	19.5	17.3	15.5	17.1	18.6	17.5	14.2		(38)
(39)		5% WB MCWBR	12.8	13.4	16.7	16.9	14.3	11.6	9.8	8.7	9.5	13.6	14.2	13.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.313	0.343	0.391	0.414	0.434	0.429	0.421	0.392	0.348	0.310	0.288		(40)
(41)		taud	2.429	2.364	2.384	2.285	2.271	2.264	2.279	2.298	2.357	2.475	2.519	2.475		(41)
(42)		Ebn at Noon	270	281	285	278	273	267	267	265	265	265	262	261		(42)
(43)		Edh at Noon	24	30	34	40	42	42	41	39	34	27	22	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	471	763	1136	1452	1688	1913	1949	1678	1359	871	544	382		(44)
(45)		RadStd	64	89	105	126	135	130	89	105	114	99	51	40		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page